Transformer: Voltage and Turns Ratio

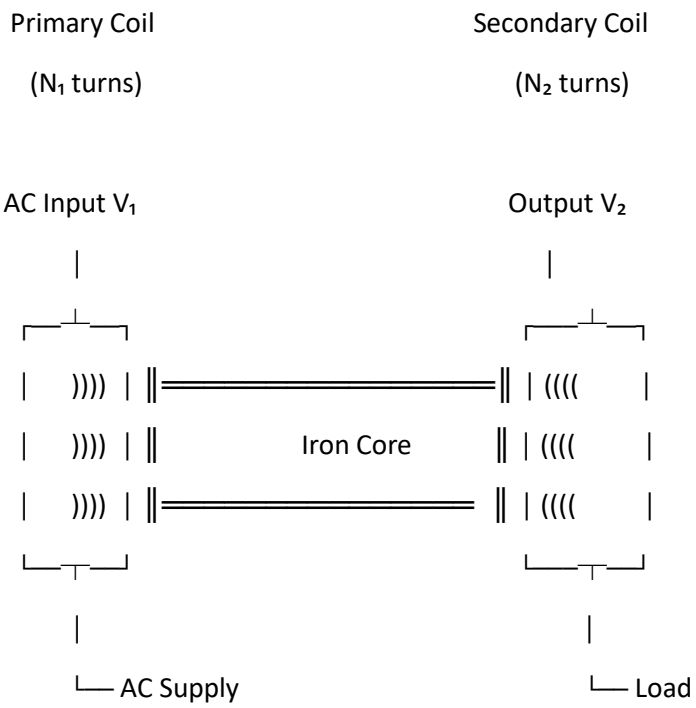
1. Definition

A transformer is an electrical device that increases or decreases AC voltage using the principle of mutual electromagnetic induction. It consists of two coils wound around a common magnetic core.

2. Principle

A transformer works on the principle of mutual induction. A changing current in the primary coil produces a changing magnetic flux, which induces an EMF in the secondary coil.

3. Diagram



4. Transformer Voltage and Turns Ratio

For an ideal transformer:

$V_1/V_2 = N_1/N_2$

Therefore:

$V_2 = V_1 \times (N_2/N_1)$

Where:

- V_1 = primary voltage (V)
- V_2 = secondary voltage (V)

- N_1 = number of turns in the primary coil
- N_2 = number of turns in the secondary coil

The turns ratio is:

Turns Ratio = N_2/N_1

Hence:

$V_2/V_1 = N_2/N_1$

5. Types of Transformers

Type	Turns relationship	Voltage
Step-up transformer	$N_2 > N_1$	$V_2 > V_1$
Step-down transformer	$N_2 < N_1$	$V_2 < V_1$
Isolation transformer	$N_2 = N_1$	$V_2 = V_1$

6. Example Calculation

Given:

- Primary voltage, $V_1 = 220$ V
- Primary turns, $N_1 = 1000$
- Secondary turns, $N_2 = 500$

Find the secondary voltage.

Formula:

$V_2 = V_1 \times (N_2/N_1)$

Calculation:

$V_2 = 220 \times (500/1000)$

$V_2 = 110$ V

Answer: The secondary voltage is 110 V, so it is a step-down transformer.

7. Current Ratio for an Ideal Transformer

For an ideal transformer, input power equals output power.

$V_1I_1 = V_2I_2$

Therefore:

$I_1/I_2 = N_2/N_1$

Equivalently:

$I_2/I_1 = N_1/N_2$

When voltage increases, current decreases, and vice versa.

8. Applications

1. Power transmission and distribution.
2. Mobile chargers and power adapters.
3. Electrical substations.
4. Voltage regulation and isolation.
5. Electronic power supplies.

9. Result

The voltage ratio of an ideal transformer is equal to its turns ratio:

$V_1/V_2 = N_1/N_2$

A step-up transformer increases voltage, whereas a step-down transformer decreases voltage.